

IRC – Incongruous Requirements Checklist

Biometric Student Attendance System — BioSec Educational Institution

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Category	Check Item	Application in This Project
Imperative	Correct priority keyword used (Shall > Must > Is required to > Will > Should)	All requirements use 'shall' as the imperative. Where weaker terms such as 'will' or 'should' appeared, these were elevated to 'shall' during regulated revision.
Continuance	No excessive use of continuance words (below, as follows, listed, in particular, supported by) that create undue complexity	Continuance language was minimal across the PS. Compound requirements such as BOR-1-MOB-02 and TR-1-BE-05 were identified and decomposed into atomic requirements to remove implied listing structures.
Ambiguity	No vague or unmeasurable terms (e.g. quickly, accurate, secure, adequate, sufficient)	Seven requirements contained vague language. BOR-1-ATT-01 used 'quickly' with no time bound. BOR-1-MOB-02 used 'accurate and up to date' with no sync interval. TR-1-BE-01 used '1000+' with no defined ceiling. All were rewritten with quantified criteria after clarification sessions.
Testability	Requirement can be verified against defined acceptance criteria	Eight requirements lacked testable criteria. BOR-1-MOB-04 defined notifications with no delivery SLA or success rate. TR-1-CPR-04 referenced 'critical system issues' with no severity classification. TR-0-02 lacked HTTPS and access scope specifications. All were given measurable acceptance conditions after stakeholder engagement.
Consistency	No conflict with another requirement in the PS	No direct contradictions were identified between requirements. Role definitions across BOR-0-01, BOR-1-MOB-02, and BOR-1-WEB-01 were reviewed and confirmed consistent after compound decomposition.
Completeness	All necessary detail, error scenarios, and edge cases are present	Seven requirements had missing detail. BOR-1-ATT-01 omitted 12 error scenarios including sensor malfunction and network outage. TR-1-BE-02 lacked MCS API endpoints, authentication, and retry logic. TR-1-CPR-01 had no lockout resolution procedures. Missing PDPA obligations for parental consent, data retention, and breach notification were also added.
Feasibility	Requirement is achievable under realistic operational conditions	Two requirements were technically infeasible. TR-1-BE-04 specified FAR of 0.001%, which is unachievable in high-traffic educational

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		environments where 0.01–0.1% is typical. TR-1-BE-02 specified 99.999% MCS uptime with no fallback, which is unrealistic without redundancy provisions. Both were revised in the technical clarification session.
Compound	Single testable concern per requirement; not combining multiple conditions with 'and'	Five requirements combined multiple concerns. BOR-1-MOB-02 combined parent access, teacher access, sync timing, and accuracy. TR-1-BE-05 combined database, encryption, backup, scan speed, FRR, FAR, web load time, and API response time. All were decomposed into atomic requirements with parent IDs retained for traceability.
PDPA Compliance	Biometric and personal data handling meets Singapore PDPA obligations	Three requirements had PDPA gaps. TR-1-CPR-03 referenced PDPA but omitted parental consent forms, data retention schedules post-graduation, data subject rights procedures (30-day response window, portability format), and breach notification mechanisms. These were added as open issues for the SRS with assigned owners and resolution dates.
External Systems	All relevant external interfaces are identified and specified	Five external systems had missing specifications. MCS integration in TR-1-BE-02 lacked API endpoints, authentication mechanism, response format, error codes, and timeout specifications. Firebase and APNS in BOR-1-MOB-04 lacked payload format and server configuration. School Management Database, Email SMTP Server, and Campus Network Infrastructure were entirely unspecified and added to the open issues register.

The IRC guided all BCIC Breakdown activities and ensured disciplined, consistent rewriting of all 27 requirements from the BioSec Project Specification.